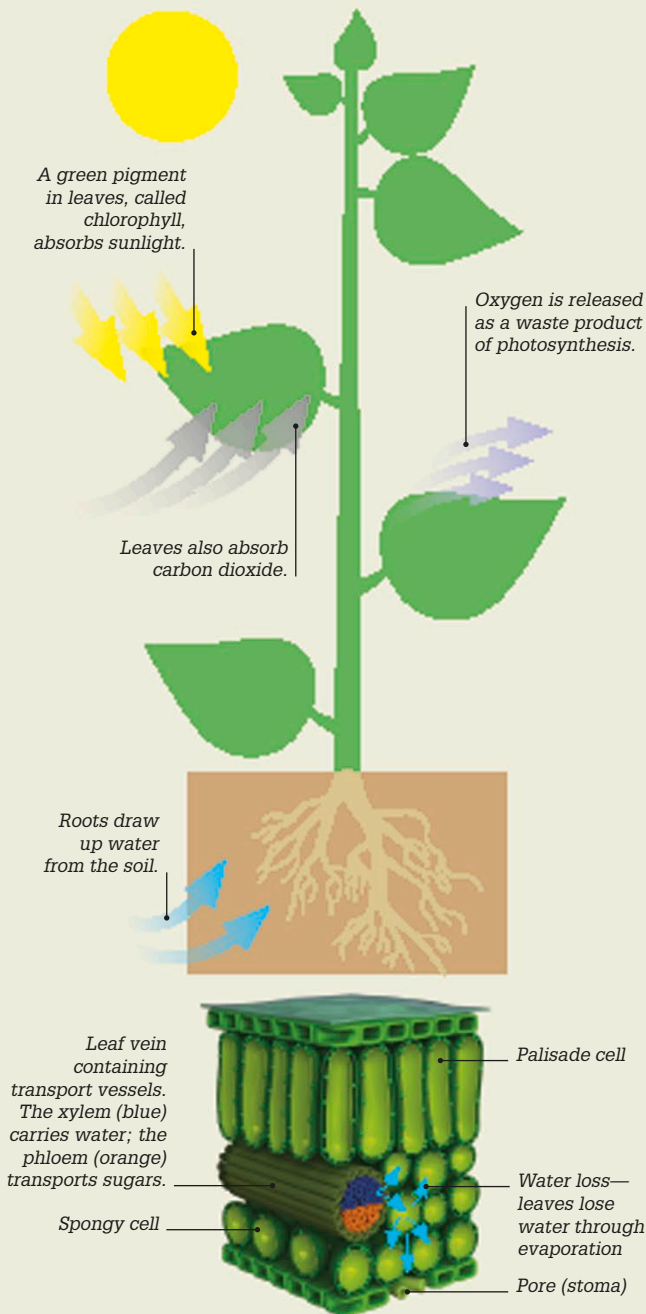


Photosynthesis

Most plants carry out photosynthesis in their leaves. In this process, green pigment called chlorophyll traps the energy in sunlight, using it to turn water and carbon dioxide into a sugar called glucose. Plants use glucose for fuel and to make other substances needed for growth. Oxygen is released as a waste product.



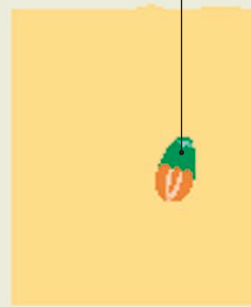
Inside a leaf

Leaves are made up of many cells. The tall palisade cells contain structures called chloroplasts, which store chlorophyll. Beneath these are transport vessels that carry water and glucose. The underside of the leaf has tiny pores (stomata), through which gases can enter and leave the leaf.

How plants grow

A plant seed contains a tiny embryo with its own supply of food. In spring, as the weather gets warmer, the embryo starts to grow and the seed sprouts (germinates). Roots grow downward to take in water and nutrients from the soil, while shoots grow upward toward the light.

Outer seed coat swells and splits.



1. Germination

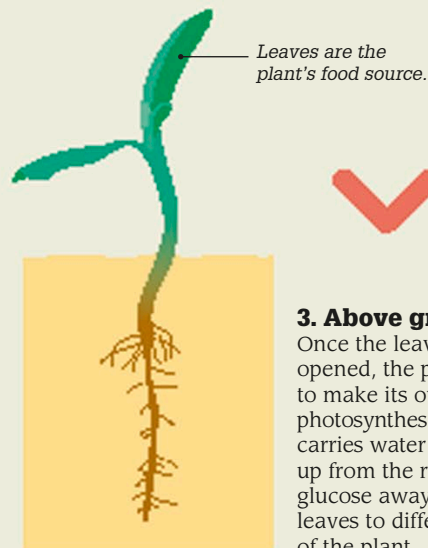
In order to germinate, the seed needs water, oxygen, and warmth. A small pore in the seed coat takes in water and the seed swells. Eventually the seed coat splits and the root and shoot emerge.

Seed sprouts and breaks through the surface of the soil.



2. New plant

The plant starts to grow beneath the ground. The roots take in water and nutrients from the soil, and help to anchor the plant in the ground. The shoot grows upward and straightens once above the ground.



3. Above ground

Once the leaves have opened, the plant can start to make its own food by photosynthesis. The stem carries water and nutrients up from the roots, and glucose away from the leaves to different parts of the plant.